SMART HOUSING WITH REMOTE CONTROL

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1. SCENARIO AND SYSTEM DESCRIPTION

Our idea concerns in realizing a smart housing IoT application that allows to gather information about external weather conditions and sense temperature and humidity of the interiors and the presence of fire inside a house.

This information will be used by our system to perform autonomous actions inside the house, that go from indoor temperature and humidity regulation to safety measures in case of domestic fire.

Also, the authenticated owner of the house can monitor remotely, using a mobile device, the status of his house and even perform himself remote control actions to change its status by turning on the air conditioning system or opening/closing windows.

2. FUNCTIONAL REQUIREMENTS

FR1: **House-owner registration**.

Description	The owner of the house needs to register him as the administrator of the system, gaining privileges to monitor house status and perform remote control actions using the mobile device.
Input	Username and Unique Key given at system's installation.
Output	Confirmation message and a House Digital Key associated to the house.

NB: House-owner registration is just a registration phase that is performed when the system is installed. It **allows having access to the House Digital Key**, useful to gain privileges about house monitoring. This registration process automatically gives administrator privileges to the user.

FR2: **Administrator authentication**.

Description	The administrator of the system can login to have remote access to the system, gaining privileges to monitor house status and to perform control actions.
Input	Username, Unique Key and House Digital Key.
Output	Confirmation message with the corresponding administrator name.

FR3: **User authentication**.

Description	The user of the system can login to have remote access to the system, gaining privileges to monitor house status.
Input	Username and House Digital Key.
Output	Confirmation message with the corresponding username.

FR4: **User interface for status monitoring**.

Description	The authenticated user sees remotely in a well organised interface, all the house information and the buttons to send control signals (if he has administrator privileges).
Input	Continuous sensor data about external and indoor temperature, windows and presence of fire.
Output	A rendered graphical interface that updates real time with the current information.

FR5: **Autonomous temperature regulation**.

Description	The system regulates autonomously the air conditioning in dependence of the external/internal temperature and if there is someone at home.
Input	Continuous information about the external/internal temperature and people presence inside the house.
Output	Control signals that regulate the air conditioning system.

FR6: **Fire detection and safety procedures**.

Description	The system gathers continuously information about the presence of fire and performs control actions in case of fatality.
Input	Continuous information about the presence of fire, status of windows and air conditioning.
Output	Control signals that put the house in a safe status and an alarm to the mobile device.

FR7: **Remote house status control**.

Description	The administrator can remotely control windows and air conditioning.
Input	Target indoor temperature that the administrator wants to reach and desired window configuration.
Output	Control signals that regulate the air conditioning or windows status.

FR8: **Temperature or windows anomaly notification**.

Description	The system sends notifications to the remote devices regarding anomalous temperature or if the windows are open when nobody is at home.
Input	Internal temperature, windows status and information about people inside the house.
Output	Notification to the remote devices.

3. NON-FUNCTIONAL REQUIREMENTS

NFR1: **Availability**. The system should always be available, especially for the fire detection, that can be considered a critical function.

NFR2: **Security and Cryptography**. The system should keep the sensitive data (as credentials) not in plaintext and the remote communications should be encrypted.

NFR3: **Ensure real-time performances**. The system must react within acceptable time limits, especially in critical situations.

NFR4: **Usability**. The system, especially the mobile application for the remote control, must be accessible and user-friendly.

NFR5: **Data Integrity**. The system must ensure that stored, transmitted and collected data remain unaltered during these phases.

NFR6: **Safety of the environment**. System automation must not interfere with the control commands produced in dangerous situations.

NFR7: **Authorization**. Only authenticated users should access the system, and different permissions should be enforced for monitoring, configuration, and control actions.